Answer the following groups

Group 1 :

A) Write the scientific term expressed by the following statements:

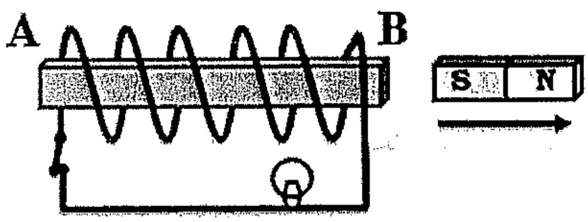
	Statement
1	**The magnetic flux density which generates a force of 1 N on a straight wire of length 1 m carrying current of intensity 1 A placed perpendicular in a uniform magnetic field** ....
2	**They are induced currents that circulate in closed paths due to the change in magnetic flux through a solid conductor associating with heating effect** ....
3	**The total number of magnetic flux lines passing through a surface and measured in weber** ....
4	**The ratio between output electric power produced from the secondary coil to the electric power given in the primary coil In the electric transformer** ....
5	**The point at which the total magnetic flux density vanishes** ....
6	**It is a phenomenon in which an induced electromotive force and also an induced current are generated in the conductor by a changing magnetic field**

B) Mention one factor only that affects each of the following:

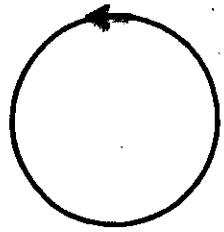
	Statement
7	The density of magnetic field at a point along the axis of solenoid carrying current. ....
8	The magnetic torque acting on a rectangular coil carrying current placed parallel in a magnetic field ....
9	The mutual induction coefficient between two coils ....
10	The induced emf in a straight wire moving normal to a magnetic field ....

Group 2 :

Put circle around the correct answer

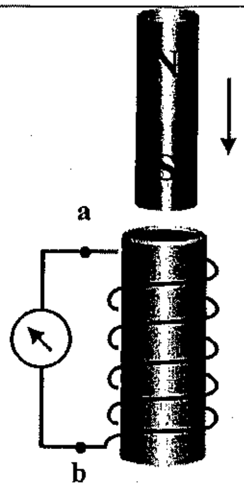
11	**The torque acting on a coil carrying current in a uniform magnetic field becomes zero when the plane of the coil is .... to the direction of the magnetic field.** a) perpendicular b) parallel c) inclined by an angle 30° d) inclined by an angle 60°
12	When a coil rotates in a magnetic field, the direction of the induced electromotive force produced in the coil is changed each ....cycle. a) $\frac{1}{4}$ b) $\frac{1}{2}$ C) 1 d) $\frac{3}{4}$
13	**When the number of turns in the dynamo's coil is doubled at constant other factors then the maximum electromotive force generated from it..............** a) Increases to the double b) decreases to half of its value c) remains constant d) Increases three times of its value
14	**In the opposite figure: When the magnet moves in the direction shown the pole formed at A is ...**  a) North b) South c) west d) east
15	**The coefficient of mutual inductance between two coils depends on <u>Except</u>.** a) The presence of an iron core inside the coil. b) The volume of the coil and the number of its turns. c) The distance separating them. d)rate of change of current

16 A metallic ring carrying a current as in the figure, the direction of the magnetic flux that produced at the center of the ring is



a) Perpendicular to the plan of the paper and inside it.
b) parallel to the plan of the paper from up to down.
c) Perpendicular to the plan of the paper and outside it.
d) parallel to the plan of the paper and inside it.

17 In the Faraday electromagnetic induction experiment shown in the diagram. Which of the following actions increases the induced emf generated across the coil at the moment of moving the magnet as shown in the diagram?



a) replacing the magnet with another with a high magnetic flux density
b) replacing the coil with another with a lower number of turns
c) increasing the speed at which the magnet moves toward the coil
d) replacing the coil with another that has a smaller cross-sectional area

18 Straight with length 2 meter is moving with a velocity 10 m/s perpendicular to uniform magnetic field with magnetic flux density 0.1 T. the induced electro-motive force generated is equal to V

a) 2
b) 1.5
c) 1
d) 0.5

19 To limit the effect of eddy currents on the iron core of the electric motor (motor) ....

a) A metal cylinder split into two equal isolated halves is required.

b) The iron core must be divided into thin sheets isolated from each other.

c) The coil must continue to rotate in the same direction.

d) A pair of solenoids is required.

20 The magnetic flux density increases at the center of a current-carrying circular coil with a decrease in ___________

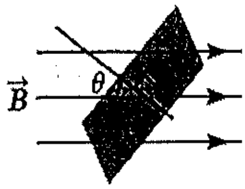
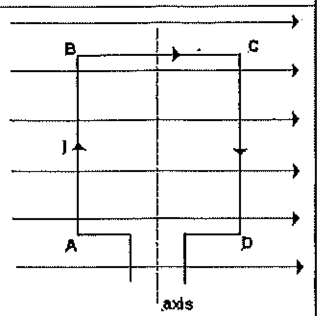
a) The number of coil turns

b) The magnetic permeability coefficient

c) The area of the face of the coil

d) The intensity of the current in the coil

Group 3:
Correct the underlined in the following statements

	Statement
21	A repulsive force exists between two parallel wires of copper each carrying current in **same** direction ....
22	The electromotive force induced in a rectangular coil rotating in a magnetic field becomes zero when the coils' plane is **parallel** to magnetic field ....
23	Magnetic field lines always emerge from the **South** Pole and terminate at the **North** Pole. ....
24	Maximum induced emf in a straight wire when the wire moving **parallel** to a magnetic field ....
25	In the following figure : The magnetic flux through a surface (Φ_m) (measured in Weber)calculated from the relation Φ_m =AB **sin θ** .... 
26	The magnetic lines of force around a straight wire carrying current are **straight lines**
27	The magnetic **force** It is the ability of medium to penetrate the magnetic flux lines
28	The density of magnetic field produced by a solenoid carrying a current at a point on its interior axis **decreases** when a soft iron bar inserted in it
29	Consider a rectangular coil of wire (ABCD) carrying a current (I) is placed in a magnetic field of flux density (B) , with its plane parallel to the field direction, as shown in the opposite figure then the force acting on Wires (AD) and (BC) is **maximum** 
30	Using switch, switch-on or switch-off the primary circuit There is an induced emf in the secondary coil only when The current in the primary coil is **constant**

Group 4:

Compare between each of the following

Ampere's right hand rule & Fleming's left hand rule in terms of the use of each

	Point of comparison	Ampere's right hand rule	Fleming's left hand rule
31 **32**	**the use**		

The physical relation used in calculating the value of the magnetic flux density at the center of a circular loop and at a point on the axis of a solenoid which the current traverse in each.

	Point of comparison	the magnetic flux density at the center of a circular loop	At point on the axis of a solenoid
33 **34**	**The physical relation used**		

Magnetic flux and Magnetic flux density

	p.o.c	Magnetic flux	Magnetic flux density
35 36	The measuring unit		

Dynamo and transformer

	Point of view	Dynamo	transformer
37 38	Function		

	the mathematical formula	The induced emf in a straight wire moving normal to a magnetic field:	the force acting on a current - carrying wire suspended at right angles to a magnetic field
39 40			

Group5:

Answer the following

	Problem
41 42	**Calculate the intensity of electric current** which passes through a circular coil of diameter 22 cm, 49 turns and produces a magnetic flux density in its center equal to 7×10^{-5} Tesla. ($\mu = 4\pi \times 10^{-7}$ Wb /A.m. ,$\pi = \frac{22}{7}$) ....
43 44	A step-down transformer is used to operate an electric lamp, which works on a potential difference of 30 volts. If an electric source of 240 volts is used & the number of tums of the primary coil is 480 turns, calculate the number of tums of the **secondary coils.** ....
45 46	An electric current of intensity 5 A flowing through a coil of self-inductance 0.001 H. If the current vanishes in 0.5 second, calculate the induced electromotive force (emf) generated in the coil. ....
47 48	If the rate of change in the intensity of the electric current passing through a coil equals 20 A/s., An electromotive force of 5 Volts is induced in the neighboring coil so that the mutual induction coefficient between the two coils is ....
49 50	A loop of cross-sectional area 0.1 m^2 and 100 turns, carrier a current of 20 A is placed parallel to a uniform magnetic field of flux density 0.5 T. Calculate the torque acting on the loop. ....

جمهورية مصر العربية
وزارة التربية والتعليم والتعليم الفني
امتحان دبلوم مدارس التكنولوجيا التطبيقية الصناعية (نظام الثلاث سنوات)
امتحان - عام experimental

الزمن: ساعتان

المادة : Physics
التخصص : جميع التخصصات الصناعية

نموذج اجابة / عام 2026 — المادة : Physics — (   )

Group 1 :

A) Write the scientific term expressed by the following statements:

1) Tesla.

2) Eddy curent
3)magnetic flux
4)effecincy of transformer
5)neutral point
6) Electromagnetic induction

B) Mention the factors affecting each of the following (one factor only)

7) The density of magnetic field at a point on interior axis of solenoid

B: the magnetic field density at the center of the circular coil **(Tesla)**

I: intensity of current passing through the coil **(Ampere)**.

L: the length of the solenoid **(meter)**.

μ: Permeability of medium

N: the number of turns of wire in the solenoid

8) magnetic field: (τ)
1) magnetic flux density **(τ α B)** (at constant other factors)
2) the current intensity **(τ α I)** (at constant other factors)
3) the area of the coil's plane **(τ α A)** (at constant other factors)
4) number of turns of the coil **(τ α N)** (at constant other factors)

9)The coefficient of mutual inductance between two coils depends on.
a) The presence of an iron core inside the coil.
b) The volume of the coil and the number of its turns.
c) The distance separating them.

10) induced emf in straight wire emf = B l v sinθ

B: magnetic flux density

L: length of wire

V: velocity of motion of wire

Group 2

<u>Put circle around the correct answer</u>

11) Perpendicular
12) ½
13) Increases to double
14) South
15) Rate of change of current
16) Perpendicular and outside it
17) a and c

a)replacing the magnet with another with a high magnetic flux density

c)increasing the speed at which the magnet moves toward the coil

18) 2V
19) The iron core must be divided into thin discs isolated from each other.

20) The area of the face of the coil

Group 3

21) Opposite

22) Normal or perpendicular

23) North –south

24) Normal

25)cos θ

26) Circular

27) Permeability

28) Increases

29)minimum or zero

30)changing

Group 4 Compare between each of the following

Ampere's right hand rule & Fleming's left hand rule in terms of the use of each

	Point of comparison	Ampere's right hand rule	Fleming's left hand rule
31 **32**	**the use**	determine the direction of magnetic field in straight wire carrying current	determine the direction of magnetic force acting on straight wire carrying current placed in a magnetic field

The physical relation used in calculating the value of the magnetic flux density at the center of a circular loop and at a point on the axis of a solenoid which the current traverse in each.

	Point of comparison	the magnetic flux density at the center of a circular loop	At point on the axis of a solenoid
33 **34**	**The physical relation used**	$B = \frac{\mu NI}{2r}$	$B = \frac{\mu NI}{L}$ OR $B = \mu n I$

	p.o.c	Magnetic flux	Magnetic flux density
35 **36**	The measuring unit	weber	tesla

	Point of view	Dynamo	transformer
37 38	Function	convert mechanical energy into electrical energy	converting low alternating voltage into high alternating voltage and vice versa

	the mathematical formula	The induced emf in a straight wire moving normal to a magnetic field:	the force acting on a current - carrying wire suspended at right angles to a magnetic field

39 40		emf = B l v	**F = B⊥ I l**

Group 5:

Answer the following

41) **42)**	$$B = \frac{\mu N I}{2r}$$ $$7 \times 10^{-5} = \frac{4\pi \times 10^{-7} \times 49 \times I}{0.22}$$ **I= 0.25A**
43) **44)**	$\frac{V_s}{V_p} = \frac{N_s}{N_p}$ $\frac{30}{240} = \frac{N_s}{480}$ Ns=60
45) **46)**	$$emf = \frac{-L\,\Delta I}{\Delta t}$$ $$emf = \frac{0.001 \times 5}{0.5}$$ **emf=0.01H**
47) **48)**	$$emf = \frac{-M\,\Delta I}{\Delta t}$$ $$5 = \frac{-M\,20}{1}$$ **M=0.25H**
49) **50)**	**The torque acting on the loop** **τ = B IAN** **τ = 0.5×20×0.1 ×100 = 100Nm**